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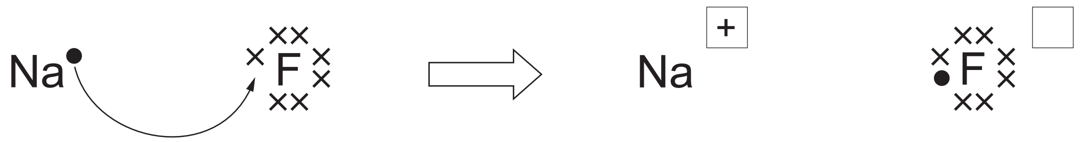
**7.** (a) The diagram shows the electronic changes that occur when sodium reacts with fluorine to form sodium fluoride. The ● and × symbols represent outer shell electrons.




| sodium atom | fluorine atom | sodium ion | fluoride ion |
|---|---|---|---|
| (2,8,1) | (2,7) | (2,8) | (......................) |

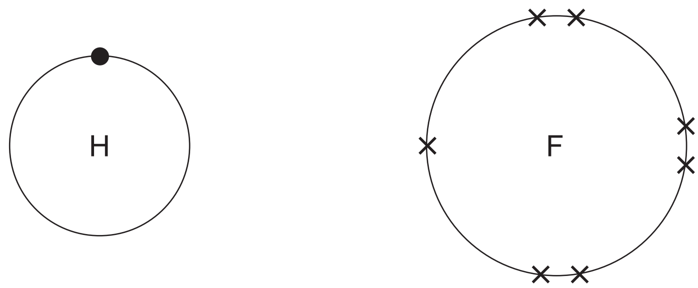
(i) **Complete the diagram** by writing the electronic structure of the fluoride ion in the brackets and its charge in the box. [2]

(ii) Complete and balance the equation for the reaction between sodium and fluorine to form sodium fluoride. [2]

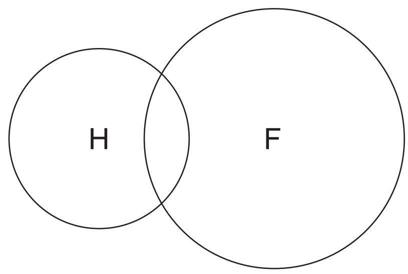
$$2Na \quad + \quad F_2 \longrightarrow \boxed{\phantom{00}} \;\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

(b) Hydrogen reacts with fluorine to form hydrogen fluoride, HF.

The diagrams show the outer shell electrons in an atom of hydrogen and an atom of fluorine.




Complete the diagram below to show the outer shell electrons in a molecule of hydrogen fluoride. [2]



(c) The equation shows the bonds which are broken and the bonds which are formed when hydrogen reacts with fluorine to form hydrogen fluoride.

$$\underset{\text{hydrogen}}{\mathrm{H{-}H}} \quad + \quad \underset{\text{fluorine}}{\mathrm{F{-}F}} \quad \longrightarrow \quad \underset{\text{hydrogen fluoride}}{\begin{matrix}\mathrm{H{-}F}\\ \mathrm{H{-}F}\end{matrix}}$$

The relevant bond energies are shown in the table.

| Bond | Bond energy (kJ) |
|---|---|
| H — H | 436 |
| F — F | 154 |
| H — F | 565 |

(i) The total energy needed to break the bonds in the H—H and F—F molecules is 590 kJ. Use the information in the table to show how this value is obtained. [1]

(ii) Use the information in the table to calculate the total energy released when the bonds in the **two** H—F molecules are formed. [1]

Energy = .................................................... kJ

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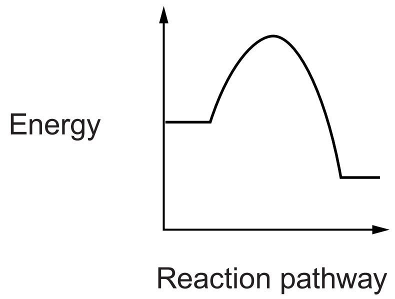
(iii) The formation of hydrogen fluoride is an exothermic reaction.

Put a **tick** (✓) in the box next to the energy profile diagram that shows an exothermic reaction. [1]

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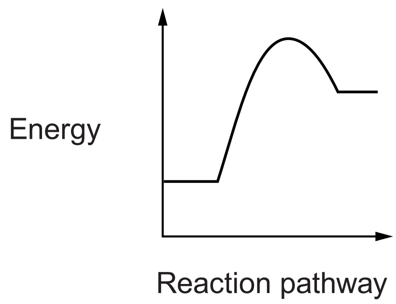


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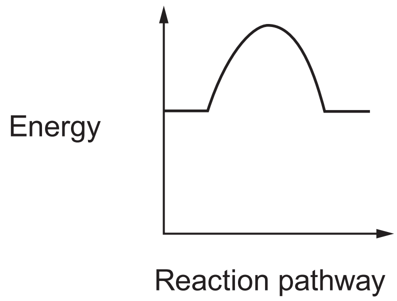


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